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10/02/2026PC-13

Version No.				
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ROLL NUMBER						

MATHEMATICS HSSC-I
(Old Curriculum)
SECTION – A (Marks 20)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اوّل لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پُر کریں۔ **Invigilator Sign.** _____

Fill the relevant bubble against each question according to curriculum: **Candidate Sign.** _____

	Question	A	B	C	D	A	B	C	D
1.	What is the multiplicative inverse of $z = 3 - 4i$?	$\frac{3}{25} + i\frac{4}{25}$	$\frac{3}{25} - i\frac{4}{25}$	$-\frac{3}{25} + i\frac{4}{25}$	$-\frac{3}{25} - i\frac{4}{25}$	○	○	○	○
2.	If $z_1 = 5 + ki$ and $z_2 = 5 - 2i$ are two equal complex numbers, the value of k is:	2	-2	3	-3	○	○	○	○
3.	For square matrices A and B, $(AB)^{-1}$ shows (in general) equality for:	$A^{-1}B^{-1}$	$B^{-1}A^{-1}$	$(BA)^{-1}$	$(AB)^{t}$	○	○	○	○
4.	The expansion of $\begin{vmatrix} 2 & 4 & 6 \\ 0 & 3 & 5 \\ 0 & 2 & 4 \end{vmatrix}$ results:	6	4	2	0	○	○	○	○
5.	The rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is:	0	1	2	3	○	○	○	○
6.	If $\underline{a} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}$ and $\underline{b} = x\hat{\imath} - 2\hat{\jmath} + 4\hat{k}$ are orthogonal (mutually perpendicular), then the value of x is:	1	-1	5	-5	○	○	○	○
7.	The geometric mean between 4 and 36 is:	9	12	18	24	○	○	○	○
8.	The sum of the first 20 natural numbers is:	200	210	220	230	○	○	○	○
9.	In how many ways can 5 different books be arranged around a circular table?	20	24	60	120	○	○	○	○
10.	What is the probability of getting a number greater than 4, when a fair die is rolled once?	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	○	○	○	○
11.	The 4th term in the expansion of $(x + y)^5$ is:	$10x^3y^2$	$5x^3y^2$	$10x^2y^3$	$5x^2y^3$	○	○	○	○

	Question	A	B	C	D	A	B	C	D
12.	The simplified form of $(x+y)^3-(x-y)^3$ is:	$6x^2y+6xy^2$	$6xy^2+2x^3$	$6x^2y-2y^3$	$6x^2y+2y^3$	○	○	○	○
13.	The graph of $y=x(x-2)(x+3)$ crosses the x-axis at:	$-1,2,3$	$0,-2,3$	$0,2,3$	$0,2,-3$	○	○	○	○
14.	The intersection of $y=x+1$ and $y=-x+5$ is:	(0,5)	(3,2)	(1,4)	(2,3)	○	○	○	○
15.	$cos3\alpha=$	$4cos^3\alpha-3cos\alpha$	$3cos\alpha-4cos^3\alpha$	$4cos\alpha-3cos^3\alpha$	$3cos^3\alpha-4cos^2\alpha$	○	○	○	○
16.	In Linear Programming, problem constraints are:	Restrictions on variables	To maximize the function	Unknowns to choose	The final solution points	○	○	○	○
17.	What is the sum of $cos(180°+\theta)$ and $cos(180°-\theta)$?	0	$2cos\theta$	$-2cos\theta$	$sin2\theta$	○	○	○	○
18.	$sin2\theta$ in terms of $tan\theta$ is:	$\frac{2tan\theta}{1-tan^2\theta}$	$\frac{1-tan^2\theta}{1+tan^2\theta}$	$\frac{1+tan^2\theta}{1-tan^2\theta}$	$\frac{2tan\theta}{1+tan^2\theta}$	○	○	○	○
19.	In a right triangle hypotenuse is 13cm with base angle $\theta=60°$. The side opposite to θ measures:	26cm	$6.5\sqrt{3}cm$	$13\sqrt{3}cm$	$\frac{6.5}{\sqrt{3}}cm$	○	○	○	○
20.	What is the domain of $y=\sin^{-1}x$?	$-\frac{\pi}{2}\le x\le\frac{\pi}{2}$	$0\le x\le 2\pi$	$0\le x\le\pi$	All real x	○	○	○	○

——1HA-II 25008-30089 (OLD)——

ROLL NUMBER						

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MATHEMATICS HSSC–I
(Old Curriculum)

Time allowed: 2:35 Hours **Total Marks Sections B and C: 80**

SECTION – B (Marks 48)

Q. 2 Solve the following Questions. **(12 x 4 = 48)**

(Use of graph paper is not necessary. Candidates can make their own grid on answer book)

(i)	If $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$ then prove that $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$	04	OR	Find the value of λ for which vectors $\underline{a} = \underline{i} + 2\underline{j} + 3\underline{k}, \underline{b} = 2\underline{i} - \underline{j} + \underline{k}$ and $\underline{c} = 3\underline{i} + 2\underline{j} + 5\underline{k}$ are coplanar.	04
(ii)	Solve for x : $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 2 \\ 1 & 2 & 3 \end{vmatrix} = 0$	04	OR	Find sum of the series $2 + 6 + 18 + 54 + \cdots + n$ terms.	04
(iii)	For the given function $f(x) = \sqrt{x+2}$, a. Find $f^{-1}(x)$. b. State domain and range of $f^{-1}(x)$.	04	OR	A cubical dice is rolled twice. a. Find the probability of getting 6 on both rolls. b. Find the probability of getting 6 on at least one roll.	04
(iv)	Solve the equation $2sin^2x - 3sinx + 1 = 0$ where $x\epsilon[0, \pi]$	04	OR	In triangle DEF, $DE = 5, EF = 7$ and $\angle D = 60°$. If $DF = k - 1$, find k using law of sines.	04
(v)	Find area of a triangle ABC with side $AB = 10$ cm, and angles $\angle A = 45°, \angle B = 75°$.	04	OR	Sketch the graph of a parabolic function $y = x^2$.	04
(vi)	Five brothers sat in a row for a group photo. Where two of them were twins. Find the number of arrangements when: a. Twins were treated distinct. b. Twins stand together (not distinct)	04	OR	The 10th term of an arithmetic sequence is 42 and the 4th term is 18. Find first term and common difference of the sequence.	04
(vii)	Find adjoint of the matrix $\begin{bmatrix} 2 & 0 & 1 \\ -1 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}$	04	OR	Insert three GMs between 5 and 80.	04
(viii)	Solve the inequality $\frac{x}{2} - 1 \leq \frac{x+3}{4} < \frac{2x+1}{3}$ and represent the solution on number line.	04	OR	Find the coefficient of x^3 in the binomial expansion of $(1 + x)^5$.	04
(ix)	Represent the solution set of a system of inequalities graphically. $x \geq 0,\ y \geq 0,\ x + y \leq 5$	04	OR	Without using calculator prove that $(sin75°cos15° + cos75°sin15°) + (cos75°cos15° - sin75°sin15°) = 1$	04
(x)	Express the complex number $\left(\frac{1+i}{2-i} \times \frac{3-i}{1+2i}\right)$ in the form $x + iy$.	04	OR	Find the maximum and minimum values of the function $f(\theta) = 7 + 4sin\theta$.	04
(xi)	If $tan\theta = \frac{3}{4}$, then find the values of $tan2\theta$ and $cos\theta$. $\left[0 < \theta < \frac{\pi}{2}\right]$	04	OR	Find angle between two vectors $\underline{a} = 2\hat{\imath}$ and $\underline{b} = \hat{\imath} + \sqrt{3}\hat{\jmath}$ using cross product method.	04
(xii)	A team of 3 students is to be selected from 7 boys and 5 girls. a. In how many ways the team has 2 boys and 1 girl? b. In how many ways the team has 1 boy and 2 girls?	04	OR	Verify by Mathematical Induction: $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$	04

SECTION – C (Marks 32)

Note: Solve the following Questions. **(4 x 8 = 32)**

(Use of graph paper is not necessary. Candidates can make their own grid on answer book)

Q.3	Solve the following system of linear equations with complex coefficients: $(1 + i)z + w = 3 + 2i,$ $z + (1 - i)w = 2 + 3i$	08	OR	Draw the graph of $y = sin\left(\frac{x}{2}\right)$ within the domain from $-\pi$ to π.	08
Q.4	Maximize $f(x, y) = 5x + 4y$ subject to the constraints $x + 2y \leq 10,\ 2x + y \leq 12,\ x + y \leq 7,\ x \geq 0,\ y \geq 0.$	08	OR	If x^4 and higher powers are neglected such that $(1 + x)^{\frac{1}{4}} + (1 - x)^{\frac{1}{4}} = a - bx^2$ find a and b.	08
Q.5	Use Cramer's rule to find the solution of the following system: $x + 2y + 3z = 14,\quad 2x - y + z = 3,\quad 3x + y - 2z = -1$	08	OR	If $\underline{a} = \underline{i} + 2\underline{j} + \underline{k}$, $\underline{b} = 2\underline{i} - \underline{j} + 3\underline{k}$ and $\underline{c} = \underline{i} + \underline{j} - 2\underline{k}$, then verify that $\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$.	08
Q.6	In a geometric progression $a_3 = 27$ and $a_5 = 243$. Find the common ratio r, first term a_1, 6th term a_6 and sum up to 6th term S_6.	08	OR	In triangle ABC, the side lengths are $a = 13$, $b = 14$ and $c = 15$. Find values of circumradius R and escribed radii r_1, r_2, r_3 of the triangle ABC.	08